

Supervised fine-tuning of decoder-only model for russian spellcheck task

Viktorov Boris

July 2026

Abstract

My project is devoted to spellcheck problem. I tested if a decoder-based model can beat encoder-decoder models that was provided by SberDevices. During tests I found that there is no label if data was typed on desktop, mobile or tablet device. I've decided to create a dataset to solve this problem. To create this dataset I've created an app to help gather this data. The dataset and models were pushed to hugging face. Project repo:https://github.com/BorisVIkVik/SpellCheck_Experiments_Project.git. Dataset app repo:https://github.com/BorisVIkVik/SpellCheck_Game_Dataset.git.

1 Introduction

In modern time we type a lot of text, so spellcheck task becomes very important to save time or not to lose face. There are articles [Martynov et al., 2024] that study this task, but they don't take account of the device that person use to type. Typing on mobile devices is a huge part of modern data, but russian open datasets that are used for spellcheck task don't label the type of device. It is happening because of gathering data that was already in the internet and you don't have any way to decide from what device it was typed. To face this problem I've made an game like app where people get sentences that they need to type and the app checks the device type that they used.

1.1 Team

Viktorov Boris - one man army.

2 Related Work

2.1 Methods and models

- [Martynov et al., 2024] is a SberDevices teams article where they proposed the SAGE framework and explored the suitability of the M2M approach

for spell checking.

- [Sharma et al., 2023]. Adobe team proposed interesting approach with ideas from RecSys field. Their approach consists of suggester and ranker. Their method gives them great improvement in latency (x10) while outperforms off-the-shelf state-of-the-art approaches in their specific domain.
- [Pal and Mustafi, 2020]. This article describes the way to correct sentences of words in Hindi that was gotten by OCR. This task has a different source of text data (OCR). They propose two-step approach that consists of error detection and error correction.

2.2 Datasets

- [Leino et al., 2024]. This dataset has a similar idea to my dataset. Their study analyses different aspects of typing data such as: WPM, ERROR_RATE, NATIVE_LANGUAGE, KEYBOARD_TYPE, , USING_APP, FINGERS, TIME_SPENT_TYPING e.t.c. , but it has data only in english and finnish.
- [Martynov et al., 2024] This work also proposed a benchmark for spellcheck task that contains multiple domains and a test/train split. Benchmark contains of RUSpellRU, MultidomainGold, MedSpellchecker and GitHub-TypoCorpusRu. (In this work I'll call it SPELLCHECK_BENCHMARK)

For comparison we've my models i've picked sage-fredt5-large. It's a model from SberDevices team that provides SOTA results on SPELLCHECK_BENCHMARK datasets.

3 Models Description

For this study I've picked Qwen2.5 and Mistral family. I've tested their different sizes to see the relationship between size of the model and performance.

4 Dataset

For this task I've decided to create my dataset. For the purpose of gathering data there was developed an app for people to type the provided message from russian classic literature. The app saves original text, typed text, anonymous user id (in dataset it's called player_id), timestamp and device type. To get the device the app checks the device that user uses to type and gives this example one of three labels: desktop, mobile or tablet. The dataset¹ was pushed to hugging face and called RU_SPELLCHECK_DEVICE.

On the Tab. 1 you can see the statistics for the mentioned dataset.

¹ BW/RU_SPELLCHECK_DEVICE.

	Train	Test
Sentences	269	269
Device types	3	
Different users	33	

Table 1: Statistics of the RU_SPELLCHECK_DEVICE.

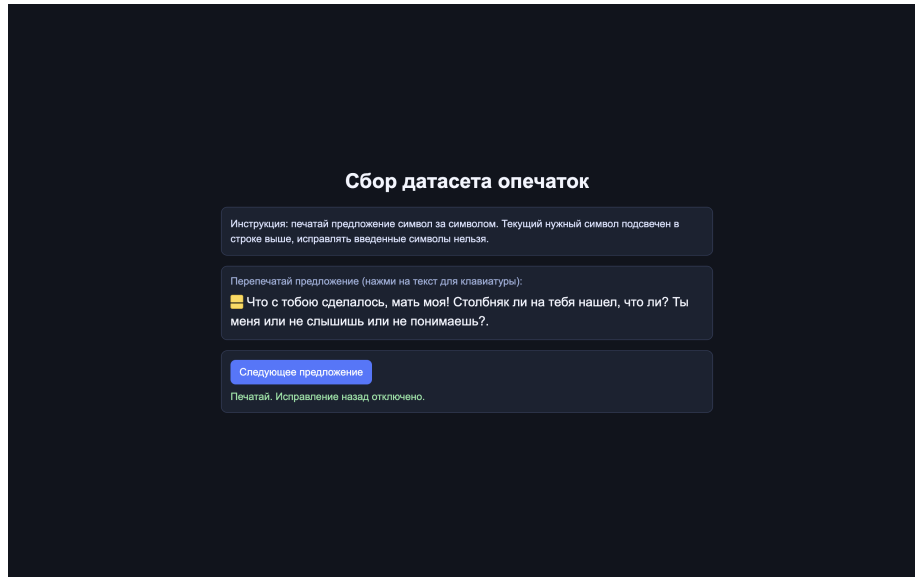


Figure 1: The app for dataset gathering.

I’ve also used `ai-forever/spellcheck_benchmark` that I will call `SPELLCHECK_BENCHMARK` and mix of `SPELLCHECK_BENCHMARK` and oversampled examples from `RU_SPELLCHECK_DEVICE`.

The comparison of T5 model results on my dataset and `SPELLCHECK_BENCHMARK` show that this dataset provides good examples that are not easy for current SOTA models.

5 Experiments

5.1 Metrics

I’ve used Errant based algorithm from sage framework to calculate the metrics. Algorithms output Precision, Recall and F1 scores that can be interpreted like the following:

Precision: one minus share of unnecessary amendments; Recall: proportion

of expected corrections; F1: famous geometric mean of aforementioned two;

5.2 Experiment Setup

Training was done in cloud.ru Finetuning service that provided H100 for SFT. Validation batch was made in google colab default GPU enviroment (T4).

5.3 Prompt optimization

Firstly, I used Genetic-Pareto (GEPA) [Agrawal et al., 2025] method to find optimal prompt for task. This approach uses reflective prompt mutation using LLM to improve current prompt. For reflective model I used gpt-oss-120b model through Cloud.ru API. Here is a table comparison for best model with base prompt and optimized.

Table 2: Comparison of SAGE metrics for the base and optimized prompts

Metric Type	Metric	Base Prompt (%)	Optimized Prompt (%)
Overall	Precision	50.69	51.90
	Recall	45.39	49.65
	F1	47.90	50.75
SPELL	Precision	51.27	54.09
	Recall	47.03	53.90
	F1	49.06	53.99
CASE	Precision	58.97	60.00
	Recall	54.76	50.00
	F1	56.79	54.55
PUNCT	Precision	51.80	53.37
	Recall	43.11	52.10
	F1	47.06	52.73

Table 3: Comparison of the base and optimized prompts.

Base Prompt	You are a Russian text proofreader. Correct spelling, punctuation, and capitalization errors in the user’s sentence. Return only the corrected text without explanations or quotation marks.
Optimized Prompt	You are a Russian-language proofreader. For every input you receive, perform only the following actions: Correct all orthographic (spelling), punctuation, and register/case mistakes. - Fix misspelled words, missing or extra letters, and wrong word forms. - Use proper Russian punctuation: commas, periods, question/exclamation marks, em-dashes (—) for dialogue, hyphens (-) only where a hyphen is required, and matching quotation marks. - Ensure correct capitalization: the first word of a sentence (or after an em-dash that starts a new sentence) must start with a capital letter; proper nouns must be capitalized; other words should be lower-case unless the context demands otherwise. - Preserve the original meaning; do not add, delete, or replace words unless it is necessary to fix an error. Output exactly the corrected text, nothing else: - No explanations, reasoning, or meta-information. - No tags such as ‘<think>’ or any other markup. - Do not surround the result with quotation marks. - Keep the original line breaks (if any) and spacing, adjusting only what is required for correct punctuation (e.g., a single space after a comma, period, dash, etc.). - If the input is already correct, output it unchanged. Examples - Input: ‘Дают приятрыз дух, увеселяют взор И вам я Красиавицы, хранят чебя в цбор.’ Output: ‘Дают приятный дух, увеселяют взор, и вам, Красиавицы, хранят себя в убор.’ - Input: ‘- Чей эоо дом? - спросил он у будувого будочника.’ Output: ‘— Чей это дом? — спросил он у углового будочника.’ - Input: ‘Тихий ветер шевелмл листьа старого дерева.’ Output: ‘Тихий ветер шевелил листьа старого дерева.’ Follow these rules for every request.

5.4 Supervised Fine-tuning

For SFT was used LoRa. Main SFT experiments were with data. I've made SFT with only RU_SPELLCHECK_DEVICE, only with 9k examples from SPELLCHECK_BENCHMARK and mixed data where was 4k examples from SPELLCHECK_BENCHMARK and 1k oversampled examples from RU_SPELLCHECK_DEVICE. We can see that best results was achieved when mixed dataset is used.

5.5 Honorable mention

New dataset give interesting opportunities in text classification task, because we can try to predict the device that the person uses. This maybe useful for improving spellcheck because typing on different device causes different error. Also in maybe useful in Criminalistics. Here are results of simple test. I've fine-tuned DeepPavlov/rubert-base-cased for predicting device label using typed text.

Table 4: Classification metrics

Метрика	Значение
Eval loss	0.73699
Eval accuracy	0.68919
Eval F1 score	0.68411
Eval precision	0.69390
Eval recall	0.68919
Epoch	5

6 Results

The best results on a test split of my dataset that I've got was made by Qwen2.5-14b model.

Table 5: Qwen2.5-14b Results

Metric Type	Precision (%)	Recall (%)	F1 (%)
Overall	51.90	49.65	50.75
SPELL	54.09	53.90	53.99
CASE	60.00	50.00	54.55
PUNCT	53.37	52.10	52.73

And here is for Sberdevices best model ai-forever/sage-fredt5-large:

Table 6: sage-fredt5-large

Metric Type	Precision (%)	Recall (%)	F1 (%)
Overall	64.42	73.73	68.76
SPELL	59.91	65.42	62.55
CASE	17.24	47.62	25.32
PUNCT	11.38	25.75	15.78

Here is a comparison table with all tested models. We can see that my models get better results in CASE and PUNCTUATION than sage-fredt5-large.

Table 7: Comparison of SAGE metrics for different models trained on different datasets. (D) - only on RU_SPELLCHECK_DEVICE, (M) - mixed data, (B) - only on SPELLCHECK_BENCHMARK

Metric Type	Metric	Qwen2.5-7B(M)	Qwen2.5-14B(M)	Ministral-8B(D)	Ministral-8B(M)	Ministral-8B(B)	sage-fredt5-large
Overall	Precision	50.16	51.90	56.90	49.85	47.81	64.42
	Recall	42.64	49.65	32.87	47.18	36.04	73.73
	F1	46.10	50.75	41.67	48.48	41.10	68.76
SPELL	Precision	48.49	54.09	56.10	52.32	46.02	59.91
	Recall	42.96	53.90	29.45	52.50	32.95	65.42
	F1	45.56	53.99	38.63	52.41	38.40	62.55
CASE	Precision	36.36	60.00	57.14	54.76	22.22	17.24
	Recall	47.62	50.00	9.52	54.76	4.76	47.62
	F1	41.24	54.55	16.33	54.76	7.84	25.32
PUNCT	Precision	1.61	53.37	32.50	50.88	3.87	11.38
	Recall	38.32	52.10	7.78	52.10	6.59	25.75
	F1	43.99	52.73	12.56	51.48	4.88	15.78

7 Conclusion

I have collected dataset with device label that is unique opensource dataset because of this. I've trained diffrenet setups of decoder-only model. My best models outperformed SberDevices model in punctuation and case metrics achieving state-of-the-art performance in this categories.

References

- [Agrawal et al., 2025] Agrawal, L. A., Tan, S., Soylu, D., Ziems, N., Khare, R., Opsahl-Ong, K., Singhvi, A., Shandilya, H., Ryan, M. J., Jiang, M., et al. (2025). Gega: Reflective prompt evolution can outperform reinforcement learning. *arXiv preprint arXiv:2507.19457*.
- [Leino et al., 2024] Leino, K., Laine, M., Kurimo, M., et al. (2024). Mobile typing with intelligent text entry: A large-scale dataset and results. *Research Square*. Preprint, Version 1.
- [Martynov et al., 2024] Martynov, N., Baushenko, M., Kozlova, A., Kolomeytseva, K., Abramov, A., and Fenogenova, A. (2024). A methodology for generative spelling correction via natural spelling errors emulation across multiple domains and languages. In *Findings of the Association for Computational Linguistics: EACL 2024*, pages 138–155.

- [Pal and Mustafi, 2020] Pal, A. and Mustafi, A. (2020). Vartani spellcheck—automatic context-sensitive spelling correction of ocr-generated hindi text using bert and levenshtein distance. *arXiv preprint arXiv:2012.07652*.
- [Sharma et al., 2023] Sharma, S., Valls-Vargas, J., King, T. H., Guerin, F., and Arora, C. (2023). Contextual multilingual spellchecker for user queries. In *Proceedings of the 46th international ACM SIGIR conference on research and development in information retrieval*, pages 3395–3399.